Benchmarking Neural Image Compression with DCT-Based Spectral Diagnostics

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Abstract

Neural image compression (NIC) models are commonly evaluated with scalar rate-distortion metrics, which do not show how a codec treats individual spatial frequencies. We propose a benchmark that probes codecs with the DCT basis matrix and single-frequency test images. We measure relative off-target energy (leakage), the fraction of reconstruction energy outside the probed frequency, together with target-frequency gain. Exact decompositions and bounds relate these measurements to reconstruction error. At operating points nearest 1 bpp, the BMSHJ, MBT, and Cheng2020 models leave 80-97% of their reconstructed high-band energy outside the target bin and have target-frequency gains of 0.02-0.23, whereas TCM and FTIC attain off-target energy in the classical-codec range. Both probe types identify behavior absent from scalar metrics: released Cheng2020-Anchor models produce extreme pre-clamp outputs on isolated single-frequency inputs. Finally, decoder-only fine-tuning with natural-image and leakage losses reduces basis off-target energy by 90-99.8% across all eight evaluated architectures while keeping the bitstream fixed, at a Kodak PSNR cost of 0.09-0.59 dB.

Code - `https://github.com/nkalmykovsk/dct_benchmark_nic`

Keywords: Neural image compression, Discrete cosine transform, Spectral analysis, Frequency response

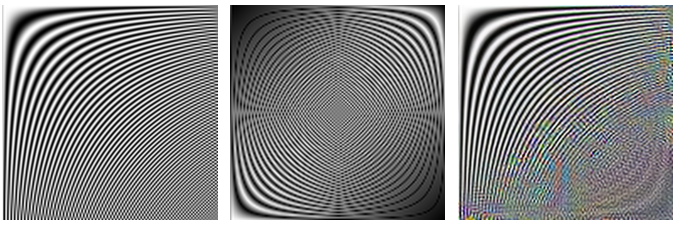
1. Introduction

Motivation and problem setting. Neural image compression (NIC) models now match or surpass traditional codecs on standard benchmarks such as Kodak [1], CLIC [2], and DIV2K [3]. Their performance is typically summarized by scalar metrics such as PSNR and MS-SSIM, which measure overall fidelity but provide little insight into how individual image structures are preserved [4, 5]. As a result, two codecs with similar rate-distortion performance may allocate compression resources very differently across spatial frequencies [6].

Most NIC frameworks [7–9] provide multiple "quality" variants corresponding to different rate-distortion trade-offs [10–12]. These variants are identified by scalar quality indices or nominal bitrate targets, which do not reveal how they differ in preserving textures, edges, and fine detail [13, 14]. Although high-frequency attenuation in learned codecs has been observed qualitatively [10], systematic measurements of its magnitude, frequency dependence, and variation across codec architectures remain limited.

This motivates two questions:

1. How do codec architectures and quality variants allocate information across different frequency bands?
2. Do higher-quality settings genuinely preserve more high-frequency detail, or merely improve global rate-distortion scores?



(a) DCT basis matrix. (b) Frequency map. (c) Cheng2020-Anchor.

Figure 1. (a) Benchmark with a normalized 128×128 DCT basis matrix. (b) Diagonal-direction frequency map, with dark = low and bright = high diagonal frequency, computed from (a). (c) Cheng2020-Anchor (q=6) reconstruction: the low-frequency upper-left region remains close to the input, whereas the high-frequency lower-right region is visibly altered.

Controlled spectral probing. We therefore introduce a DCT-basis diagnostic benchmark for measuring frequency-dependent codec output. It uses two probes: a composite image formed from the full DCT basis, which the codec processes jointly as one stress image, and isolated single-frequency inputs that place one component per image. From codec reconstructions we derive a frequency-response matrix and diagnostics for target-frequency gain and the distribution of output energy across frequencies. Only the isolated probes admit an independent per-frequency interpretation, in the spirit of sine-sweep testing but with the caveat that practical codecs are nonlinear and content adaptive; the composite basis is a joint stress test rather than a set of independent measurements.

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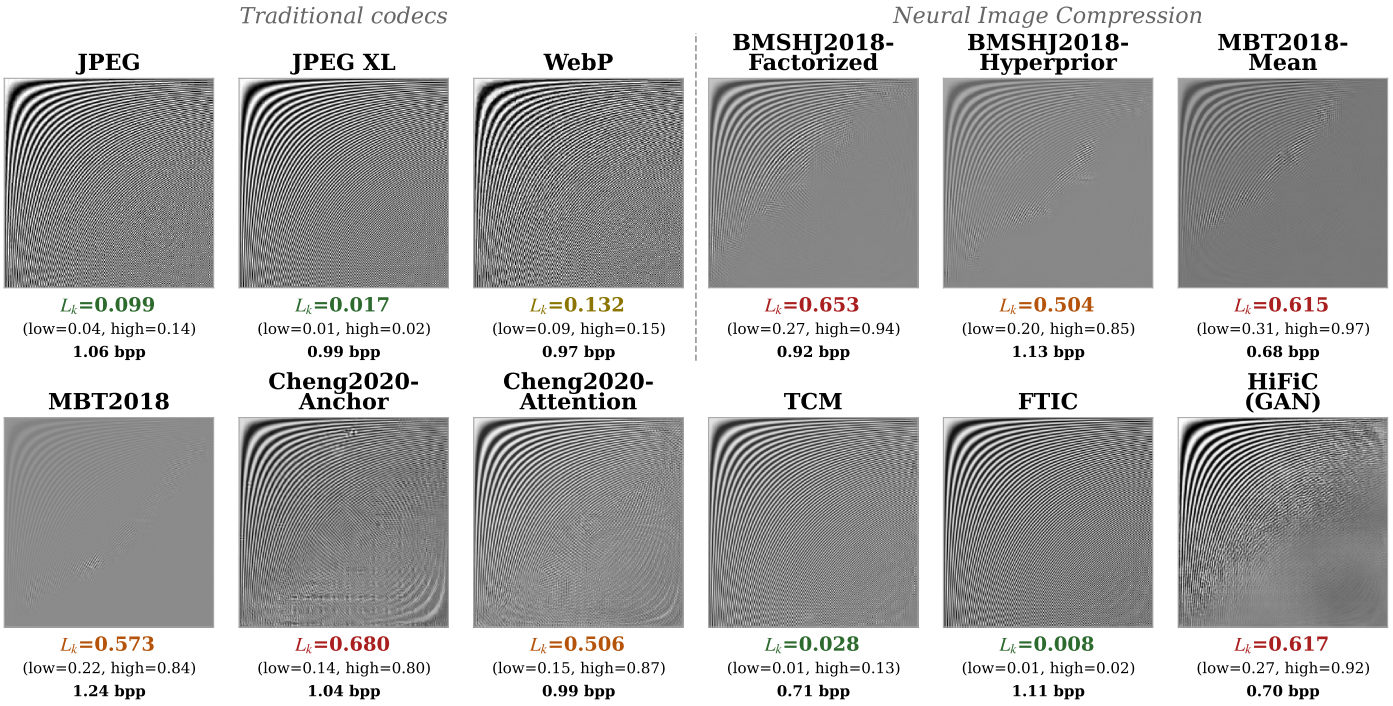


Figure 2. DCT-basis response at the nearest available operating point to ≈1.0 bpp (per-panel bpp shown), 256 × 256, for eleven codecs plus the GAN-based HiFiC codec (bottom-right, at its highest fixed rate, 0.70 bpp; best viewed zoomed in). The three classical codecs are left of the dashed line; the learned codecs follow. Under each panel: median leakage L_k (↓ better, colored green→red by magnitude), its *low*/*high*-band values, and the measured bitrate. At these nearby rates the high-frequency leakage gap between learned and classical codecs persists (L_{high}: 0.02-0.15 for traditional codecs vs. 0.80-0.97 for the BMSHJ/MBT/Cheng families); TCM and FTIC fall in the classical range under this diagnostic. HiFiC (L_k=0.62) shows structured content in the region with high measured leakage rather than the smooth blur of the MSE-trained decoders.

Practical scope. The controlled-probe evaluation complements natural-image evaluation. Conventional metrics quantify aggregate distortion; the controlled probes localize changes in the measured spectral response (Figs. 2 and 3), while their relation to natural-image error is summarized only by descriptive alignment statistics (Table 2). The observed failures are content-dependent rather than universal. In the selected Urban100 [15] and DTD [16] examples, pronounced artifacts occur on repetitive high-frequency structures such as facade slats, lace, fine line patterns, zigzags, and meshes, whereas an irregular texture in the same gallery is much less affected (Suppl. Figs. S5-S6). Such localized failures can be diluted by dataset-level averages. Controlled probes are therefore useful for codec and checkpoint stress testing, regression testing across operating points, and decoder regularization; they are not intended to predict the error of an individual natural image.

Contributions. We make three contributions:

- **Controlled DCT benchmark:** We combine a basis DCT matrix image and isolated single-frequency probes, with relative off-target energy and gain as its primary outputs (Fig. 3); at nearby measured rates, the older BMSHJ, MBT, and Cheng2020 models have high-band leakage of 0.80-0.97 and gains of 0.02-0.23, whereas TCM and FTIC fall in the classical-codec range under this diagnostic (Fig. 2, Table 1, Fig. 4);
- **Error analysis and failure diagnosis:** We derive exact error decompositions and leakage-error bounds (Eqs. (5)-(9)) and use isolated probes to identify a frequency-specific instability in the released Cheng2020-Anchor checkpoints (Fig. 5); two tested input-independent response models do not transfer quantitatively to natural images in the evaluated setting, motivating a descriptive alignment summary over 115 images (Table 2);
- **Decoder-only leakage regularization:** Across all eight evaluated architectures, fine-tuning with natural-image and leakage losses reduces basis off-target energy by 90-99.8% while keeping the bitstream fixed, at a Kodak PSNR cost of 0.09-0.59 dB (Table 3, Fig. 7).

2. Related Work

Transform coding and learned-codec evaluation. Classical codecs make frequency allocation explicit through transforms and coefficient quantization, with the DCT underlying JPEG and related coding standards [17–19], and learned models have been applied in the DCT domain itself, e.g. for artifact removal [20]. Learned image compression instead optimizes nonlinear transforms and entropy models jointly, progressing from factorized models [7] and hyperprior models [21] to autoregressive priors [22], attention mechanisms [8, 23], space-channel context models [11], and hybrid transformer-CNN architectures [9]. The approach has been standardized as JPEG AI [24]. Surveys and benchmarks compare these systems primarily through rate-distortion and perceptual metrics on natural-image datasets

[4, 5, 25]. Such evaluation establishes overall coding efficiency but does not identify the response to a particular input frequency. We use the DCT basis as a controlled stimulus rather than as a codec transform, and retain natural-image metrics as a separate evaluation axis.

Spectral bias and codec diagnostics. Spectral-bias studies show that neural networks tend to learn low-frequency *functions* [26], not the frequency response of a rate-constrained codec. Lieberman et al. [6] used Fourier summaries of reconstruction residuals to study NIC generalization and robustness. This is an *a posteriori* analysis of natural or corrupted images, whereas our protocol controls the input spectrum. The composite basis yields a response matrix, while isolated inputs provide independent per-frequency measurements (Sec. 3.3); target gain and off-target energy are measured separately. Natural-image spectra and controlled probes are complementary; we do not assume quantitative transfer between them (Sec. 5.4). The contribution is the controlled-probe protocol and its error decomposition.

Frequency-aware codec design. Frequency-aware codecs use explicit decomposition [27], FTIC's directional attention [28], WeConvene's wavelet convolution and entropy model [29], auxiliary wavelet transforms [30], content-adaptive transforms [31], or multiscale regularization of energy dispersion and aliasing [32]. These methods modify the architecture or training objective to improve rate-distortion performance. Our model-agnostic benchmark instead measures target-frequency gain and off-target redistribution in trained codecs under the same controlled inputs. The TCM and FTIC results are diagnostic comparisons, not causal evidence about individual architectural components.

Robustness and distribution shift. NIC robustness has been studied through common corruptions [6], adversarial rate or distortion attacks and fine-tuning defenses [33–35], robustness benchmarks [36], and instability under successive compression [37]. These works concern distribution shift, repeated coding, or deliberately optimized perturbations. Our isolated gratings are instead structured stress stimuli with known spectral support. The observed Cheng2020-Anchor failure is reported only for the released checkpoint family (Fig. 5); the experiment neither isolates an architectural cause nor characterizes behavior on deployment data.

Perceptual and generative compression. Generative compression methods address the rate-distortion-perception tradeoff [14]. Examples include HiFiC [10], implicit local-likelihood models [38], and conditional diffusion decoders [39]. Some spectral-consistency objectives use high-frequency discriminators [40]. Perceptual or distributional fidelity does not directly measure target-frequency preservation. We include HiFiC as a perceptual reference and report its relative off-target energy. We do not generalize its measured behavior to other generative codecs.

3. Methodology

3.1. DCT-Basis Probe

The orthonormal DCT basis can be assembled columnwise as an $n \times n$ composite image $\mathbf{D}$ (Fig. 1(a)). Frequencies increase from the upper-left to the lower-right region. Each column is a single-frequency pattern, but the codec processes all columns jointly as one image. Independent per-frequency analysis uses the isolated probes of Sec. 3.3. Before compression,

$$\mathbf{D}^\top \mathbf{D} = \mathbf{I}$$

so each column is localized to one DCT bin. Compression may change the target coefficient and move output energy to other bins; target-frequency gain and relative off-target energy quantify these effects separately (Table 1). Notation is collected in Suppl. Table S1.

The assembled basis also forms diagonal chirp trajectories through Nyquist folding of its quadratic phase (Fig. 1(b)); their derivation and interpretation are given in Suppl. Sec. B.2.

3.2. Response Matrix and Metrics

Let d_k be column k of $\mathbf{D}$ and $\widehat{d_k}$ the corresponding column after joint compression and reconstruction of the composite image. We apply the 1-D DCT $\mathcal{D}(\cdot)$ to each reconstructed column and, for positive output energy, define the *frequency-response matrix*

$$R_{i,k} = \frac{|\mathcal{D}(\widehat{d_k})_i|^2}{\sum_{j=0}^{n-1} |\mathcal{D}(\widehat{d_k})_j|^2}, \tag{1}$$

where $R \in \mathbb{R}^{n \times n}$. Each column of R sums to one: $R_{k,k}$ is the output-energy fraction in the target bin, and $R_{i,k}$ for $i \neq k$ is the fraction assigned to bin i. Lossless reconstruction gives $R = I$. Equation (1) applies to positive-energy columns, as in all evaluated runs. The implementation uses $\epsilon = 10^{-12}$ in numerical denominators and assigns $R_{:,k} = 0$ and $L_k = 1$ to a zero-output column; exact identities in Suppl. Sec. B.4 use the positive-energy definition above.

The columnwise transform preserves the index k of each target pattern. A 2-D DCT would instead describe the joint spatial layout of the composite image and would not yield the same column-indexed response.

We use two primary diagnostics, L_k and g_k; auxiliary quantities describe the response shape.

- **Relative off-target energy (leakage):** Fraction of the reconstructed column's energy outside the target bin,

$$L_k = 1 - R_{k,k} \in [0, 1], \tag{2}$$

  which measures normalized redistribution and is unchanged by uniform amplitude scaling.

- **Target-frequency gain:** Signed target-bin coefficient, $g_k = \mathcal{D}(\widehat{d_k})_k$, for a unit input coefficient. It measures target amplitude. If $\widehat{d_k} = \alpha d_k$, then $L_k = 0$ but $g_k = \alpha$; reporting both separates attenuation from redistribution.

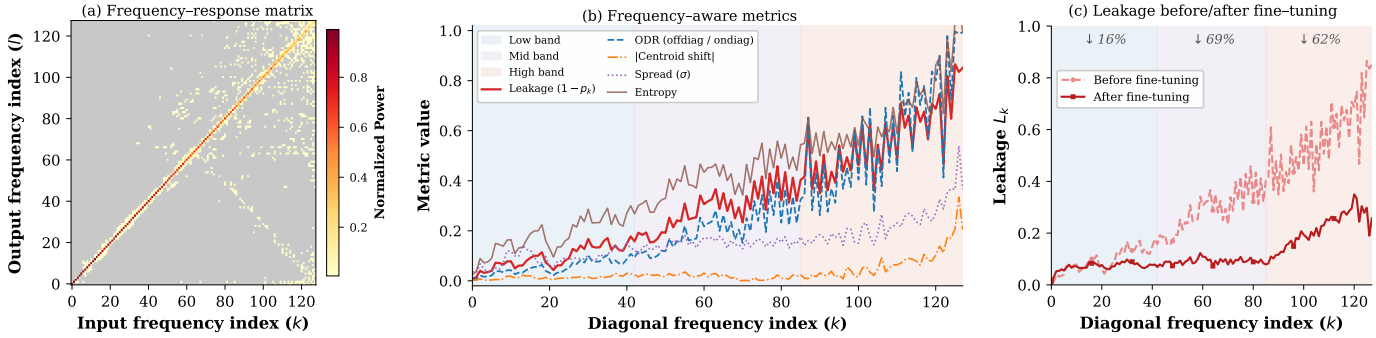


Figure 3. Frequency-domain analysis of Cheng2020-Anchor on a 128×128 DCT basis ($q = 6$). (a) The frequency-response matrix shows strong off-target redistribution in high-frequency components (top-right) and near-perfect target-bin purity in low-frequency components (bottom-left). (b) Frequency-aware metrics quantify these effects across low, mid, and high bands (in the panel legend, p_k denotes the retention $R_{k,k}$, so leakage is $1 - p_k$). (c) Leakage L_k before vs. after decoder-only fine-tuning with the proposed leakage loss (900 steps; protocol in Sec. 5.5); percentages indicate per-band reduction.

- **Off-Diagonal Ratio (ODR):** Bounded ratio of off-target to target energy,

$$\mathrm{ODR}_k^{\mathrm{norm}} = \tanh\left[\frac{\sum_{i\neq k} R_{i,k}}{2(R_{k,k}+\epsilon)}\right],$$

  used only as a display-scale reparameterization of L_k.

- **Centroid Shift:** Normalized displacement of the output-energy centroid from the target index,

$$\mu_k = \sum_i i\, R_{i,k}, \quad \Delta_k = \frac{|\mu_k - k|}{\max(k,\ n{-}1{-}k)}.$$

- **Spread:** Standard deviation about μ_k, normalized by the distance to the farther band edge,

$$\sigma_k^{\mathrm{norm}} = \frac{\sqrt{\sum_i (i-\mu_k)^2 R_{i,k}}}{\max(\mu_k,\ n{-}1{-}\mu_k)+\epsilon}.$$

- **Entropy:** Normalized Shannon entropy of the output-energy distribution,

$$H_k^{\mathrm{norm}} = \frac{-1}{\log n}\sum_i R_{i,k}\log(R_{i,k}+\epsilon),$$

Low Δ_k, σ_k^{norm}, and H_k^{norm} indicate limited displacement, dispersion, and spectral disorder, respectively. Pooled Spearman correlations with L_k are 0.98 for entropy and 0.92 for spread. Quantitative comparisons therefore use L_k and g_k, while the auxiliary metrics interpret Fig. 3(b). Formal identities and pooled analyses are given in Suppl. Sec. B.4.

3.3. Isolated Single-Frequency Probes

The basis-matrix probe processes all n basis patterns jointly. For independent per-frequency measurements, we instead place one 2-D DCT component in each test image. Let $\mathbf{d}_k \in \mathbb{R}^n$ be column k of $\mathbf{D}$ and $\mathbf{1} \in \mathbb{R}^n$ the all-ones vector. We define two unit-energy, rank-1 probes:

$$\mathbf{X}_k^{(1)} = n^{-1/2}\,\mathbf{d}_k\mathbf{1}^\top \in \mathbb{R}^{n\times n}, \tag{3}$$

$$\mathbf{X}_k^{(2)} = \mathbf{d}_k\mathbf{d}_k^\top \in \mathbb{R}^{n\times n}, \tag{4}$$

with $\|\mathbf{X}_k^{(1)}\|_F = \|\mathbf{X}_k^{(2)}\|_F = 1$; both are shown in Fig. S1.

For codec input, each probe is embedded into $[0, 1]$ by the centered affine map $x \mapsto \frac{1}{2} + sx$, using one fixed scale s for all k, replicated over three channels, and mapped back after decoding. We do not use min-max normalization because it introduces a k-dependent DC offset and changes the measured leakage.

Single-bin spectra. Let $\mathcal{D}_2(\mathbf{X}) = \mathbf{D}^\top\mathbf{X}\mathbf{D}$ denote the 2-D DCT. The probes satisfy

$$\mathcal{D}_2(\mathbf{X}_k^{(1)}) = \mathbf{e}_k\mathbf{e}_0^\top, \qquad \mathcal{D}_2(\mathbf{X}_k^{(2)}) = \mathbf{e}_k\mathbf{e}_k^\top.$$

Thus $\mathbf{X}_k^{(1)}$ and $\mathbf{X}_k^{(2)}$ occupy the single unit-amplitude bins $(k, 0)$ and (k, k), respectively. The proof and a controlled attenuation check are given in Suppl. Sec. B.3.

$\mathbf{X}_k^{(1)}$ varies along one spatial axis, whereas $\mathbf{X}_k^{(2)}$ varies along both. Their responses are analyzed separately; their error ratio is not used as an orientation measure. After inverse affine mapping, reconstructed energy outside the target bin is attributed to the codec response.

3.4. Exact Error Decomposition and Leakage Bounds

Let $\mathbf{X}$ be either probe, with unit coefficient in target bin (a, b), and let $\widehat{\mathbf{G}} = \mathcal{D}_2(\widehat{\mathbf{X}})$ be the 2-D DCT of its reconstruction. Define the *target-frequency gain* $g_k = \widehat{G}_{a,b}$ and *off-bin energy* $E_k^{\mathrm{off}} = \sum_{(i,j)\neq(a,b)} \widehat{G}_{i,j}^2$. By Parseval's identity, every reconstruction satisfies

$$\|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2 \;=\; (1 - g_k)^2 \;+\; E_k^{\mathrm{off}}. \tag{5}$$

The two terms are the target-coefficient error and the energy outside the target bin.

Define the single-frequency *energy leakage* λ_k for $\widehat{\mathbf{G}} \neq 0$ as

$$\lambda_k = \frac{E_k^{\mathrm{off}}}{\|\widehat{\mathbf{G}}\|_F^2} = 1 - \frac{g_k^2}{\|\widehat{\mathbf{G}}\|_F^2}.$$

This is the isolated-probe analogue of L_k in Eq. (2); for an all-zero reconstruction, we set $\lambda_k = 1$. With this convention, every reconstruction satisfies

$$\lambda_k \;\le\; \|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2. \tag{6}$$

If $\|\widehat{\mathbf{G}}\|_F = 1$ and $g_k \geq 0$, then

$$\|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2 \;\leq\; 2\lambda_k. \tag{7}$$

The lower bound requires no output-energy normalization; the upper bound does. Without this constraint, no finite upper bound can depend on λ_k alone. Proofs, equality conditions, and optimality of the constants are given in Suppl. Sec. B.4.

3.5. Full-Basis Error Bounds

Exact relations among the auxiliary response metrics are given in Suppl. Sec. B.4. Here we apply the single-frequency bounds columnwise to the basis-matrix probe. Let $\widehat{\mathbf{D}} = [\widehat{\mathbf{d}}_0, \ldots, \widehat{\mathbf{d}}_{n-1}]$ be its reconstruction and $\bar{L} = n^{-1} \sum_k L_k$ the mean leakage. If every reconstructed column has positive energy, then

$$\tfrac{1}{n} \|\widehat{\mathbf{D}} - \mathbf{D}\|_F^2 \;\geq\; \bar{L}, \tag{8}$$

without assuming energy preservation. If, in addition, each column satisfies $\|\mathcal{D}(\widehat{\mathbf{d}}_k)\|_2 = 1$ and $\mathcal{D}(\widehat{\mathbf{d}}_k)_k \geq 0$, then

$$\tfrac{1}{n} \|\widehat{\mathbf{D}} - \mathbf{D}\|_F^2 \;\leq\; 2\bar{L}. \tag{9}$$

Thus Eq. (8) bounds mean per-column squared error without output-energy preservation, whereas Eq. (9) requires the stated normalization and sign conditions. Proofs are given in Suppl. Sec. B.4.

Weighting the profiles by a contrast-sensitivity function leaves the codec ranking essentially unchanged (ρ_s=0.991; Suppl. Sec. C.1).

3.6. Natural-Image Spectral Alignment

Leakage L_k (2) is measured on synthetic inputs. To compare its frequency profile with natural-image distortion, we map both to a common radial-frequency axis. For image X, reconstruction $\widehat{X}$, and error $e = X - \widehat{X}$, let $P_X(f)$ and $P_e(f)$ be their 2-D DCT powers averaged within N_b radial rings. Define the *radial distortion fraction*

$$\rho(f) = \frac{P_e(f)}{P_X(f) + P_e(f) + \epsilon}, \qquad \rho(f) \in [0, 1]. \tag{10}$$

We linearly interpolate L_k at $k(f) = f(n-1)/f_{\max}$, where $f_{\max}$ is the largest radial ring frequency, clamped to $[0, n-1]$, to obtain $L^{\mathrm{r}}(f)$. This mapping assumes isotropy and serves only as a radial proxy. We define

$$\tilde{\mathcal{L}}(X, \widehat{X}) = \frac{1}{N_b} \sum_{f=0}^{N_b-1} \rho(f)\, L^{\mathrm{r}}(f). \tag{11}$$

For $\bar{\rho} = N_b^{-1} \sum_f \rho(f) > 0$, the ratio $\tilde{\mathcal{L}}/\bar{\rho}$ is the ρ-weighted mean of $L^{\mathrm{r}}(f)$: it is high when measured distortion concentrates at frequencies where the mapped synthetic profile leaks, and low when the two profiles do not overlap. It summarizes spectral alignment only; it is neither a distortion predictor nor a causal attribution. Transfer results and limitations are reported in Sec. 5.4.

3.7. Leakage-Regularized Fine-Tuning Loss

We fine-tune the decoder using two loss terms. The natural-image term is the mean pixel MSE over a batch $\mathcal{B}$ of image patches,

$$\mathcal{L}_{\mathrm{dist}} = \frac{1}{|\mathcal{B}|} \sum_{\mathbf{X} \in \mathcal{B}} \mathrm{MSE}(\widehat{\mathbf{X}}, \mathbf{X}). \tag{12a}$$

The frequency term is computed on a fixed composite DCT-basis probe. For its reconstruction $\widehat{\mathbf{D}}$, we compute R from Eq. (1), set $L_k = 1 - R_{k,k}$, and define

$$\mathcal{L}_{\mathrm{freq}} = \sum_{k=0}^{n-1} w_k L_k. \tag{12b}$$

We set $k_0 = \lfloor 0.33n \rfloor$, so this term acts on approximately the upper two-thirds of the frequency indices. Within the active band, the weights increase with frequency:

$$\widetilde{w}_k = \begin{cases} 0, & k < k_0, \\ (k/(n-1))^p + 0.2, & k \geq k_0, \end{cases} \qquad w_k = \frac{\widetilde{w}_k}{\sum_j \widetilde{w}_j}. \tag{13}$$

The constant 0.2 provides a fixed baseline weight within the active band. We use p=3 for leakage-regularized fine-tuning.

Because $\mathcal{L}_{\mathrm{freq}}$ targets only $k \geq k_0$, two additional terms constrain the low-frequency response. Let $R^{(0)}$ denote the response matrix before fine-tuning, and let $\mathbf{D}_{<k_0}$ and $\widehat{\mathbf{D}}_{<k_0}$ denote the target and reconstructed submatrices formed by the first k_0 columns, respectively:

$$\mathcal{G}_{\mathrm{pur}} = \frac{1}{k_0} \sum_{k<k_0} \max(0, R^{(0)}_{k,k} - R_{k,k} - 0.02), \tag{14}$$

$$\mathcal{G}_{\mathrm{mse}} = \mathrm{MSE}\Big(\widehat{\mathbf{D}}_{<k_0}, \mathbf{D}_{<k_0}\Big). \tag{15}$$

$\mathcal{G}_{\mathrm{pur}}$ penalizes low-band purity decreases beyond a fixed 0.02 margin, and $\mathcal{G}_{\mathrm{mse}}$ penalizes low-band reconstruction error. The full objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{dist}} + \lambda_{\mathrm{freq}} \mathcal{L}_{\mathrm{freq}} + 0.12\mathcal{G}_{\mathrm{pur}} + 0.02\mathcal{G}_{\mathrm{mse}}. \tag{16}$$

The MSE-only control uses only $\mathcal{L}_{\mathrm{dist}}$ in Eq. (12a). Leakage-regularized fine-tuning uses the full objective in Eq. (16). We select $\lambda_{\mathrm{freq}} \in \{0.003, 0.01, 0.03\}$ using the held-out DIV2K monitor (Sec. 5.5; Table 3). In both protocols, only the decoder is updated; the encoder, hyperprior, and entropy model remain fixed, so the bitstream is unchanged.

4. Experimental Protocol

We evaluate frequency-aware rate-distortion performance using a unified pipeline across classical codecs (JPEG, JPEG XL, WebP) and learned codecs: six models from CompressAI [41] (BMSHJ2018-Factorized, BMSHJ2018-Hyperprior [21], MBT2018-Mean, MBT2018 [22], Cheng2020-Anchor, Cheng2020-Attention [8]), as well as TCM [9] and FTIC [28]. As a perceptual anchor we additionally profile the GAN-based HiFiC [10] at its three released operating points (Figs. 2 and 4); its rates are fixed by the released checkpoints, so it enters the anchored comparisons rather than the nearest-rate comparisons. For each tested image size

($n \in \{64, 128, 256, 512, 1024\}$), we generate normalized three-channel DCT basis images and compress them at multiple quality levels or bitrates. The decompressed outputs are mapped back to the basis scale and analyzed to obtain the frequency-response matrix R (Section 3.2) and the proposed metrics. The basis figures use each codec's available setting nearest 1.0 bpp (0.68-1.24 bpp; HiFiC is fixed at 0.70 bpp; Fig. 2). For natural images, per-dataset settings yield codec-level mean rates of 0.82-0.98 bpp (Table 2). All stimuli are achromatic (the basis replicated across RGB); chromatic probing, relevant given the visibly chromatic artifacts of Figs. 1(c) and 6(b) and the chroma subsampling of classical codecs, is left to future work. Full reproducibility details are given in Suppl. Sec. F.

5. Results and Discussion

5.1. Spectral Allocation Across Codecs

Figure 2 compares eleven codecs at their available settings nearest 1.0 bpp and HiFiC at its highest fixed rate. The separation in median leakage persists across the evaluated rate sweep (Fig. 4).

JPEG, JPEG XL, and WebP have high-band relative off-target energy $L_{\text{high}} \leq 0.15$. The BMSHJ/MBT/Cheng families reach L_{high}=0.80-0.97 and differ mainly in their low- and mid-band response. TCM and FTIC have overall median leakage $L_k \leq 0.03$ and fall in the classical-codec range. Cheng2020-Anchor has the highest median leakage: L_k=0.68 at 256×256 (q=1; Fig. 2) and 0.73 at 512×512 (q=6; Table 2).

Table 1 separates target-frequency gain from off-target redistribution. The BMSHJ, MBT, and Cheng2020 models have high-band gains of 0.02-0.23, compared with 0.78 and 0.97 for TCM and FTIC. Thus, high off-target energy co-occurs with low target gain at these settings, but the two quantities remain distinct.

Rate selection on the basis yields harsher classical-codec settings than rate selection on natural images. Figure 2 and Table 2 therefore state their protocols separately; their classical-codec leakage values are not directly interchangeable. TCM uses λ=0.05 (p=128) in Table 2 and its p=64 variant in the basis-rate comparisons.

5.2. Single-Frequency Response

Composite-basis leakage L_k measures columns processed jointly, whereas λ_k is obtained from isolated inputs. With decoder outputs clamped to $[0, 1]$ at settings nearest 1 bpp, the profiles agree on the $\mathbf{X}^{(2)}$ sweep: Spearman's ρ_s is 0.90-0.97 for the eight learned codecs and 0.89 for JPEG. Correlation is lower for JPEG XL (ρ_s=0.37), where leakage is near the numerical floor (median $\lambda_k \approx 6 \cdot 10^{-3}$), and for WebP on the $\mathbf{X}^{(1)}$ gratings (ρ_s=0.59), where the low-contrast gratings are strongly suppressed at this setting (median $\lambda_k \approx 0.6$). For Cheng2020-Anchor at its highest quality, removing the clamp reduces the correlation to approximately zero because extreme outputs dominate the profile (Sec. 5.3).

The two constructions $\mathbf{X}^{(1)}=(k, 0)$ and $\mathbf{X}^{(2)}=(k, k)$ differ in contrast, radial frequency, and spatial structure. Their responses are therefore reported as complementary probe geometries and are not used to claim an isolated orientation effect.

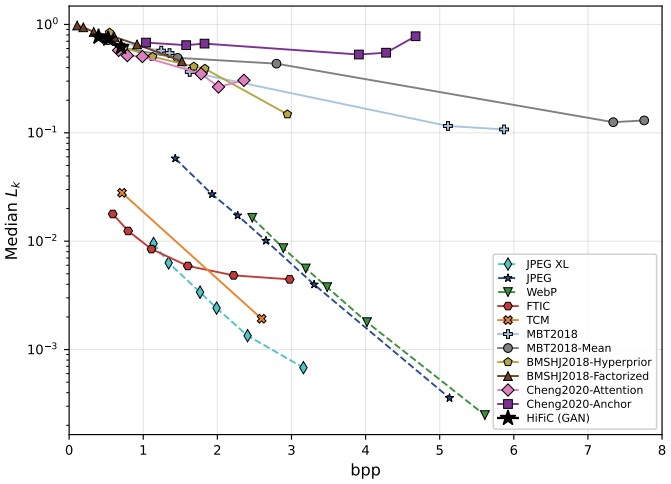


Figure 4. DCT rate-leakage comparison at 256 × 256. Traditional codecs (dashed) achieve lower median leakage across all bitrates than the older BMSHJ/MBT/Cheng models (solid), while TCM and FTIC approach classical levels. The GAN-based HiFiC is shown at its three fixed operating points (black stars).

Table 1. Composite-basis response of learned codecs at their settings nearest 1.0 bpp (256×256). Values are means over the top-quarter band ($k \geq 3n/4$); best values are bold. Classical codecs are not included.

Codec	bpp	$L_{\text{high}} \downarrow$	gain $g_k \uparrow$
BMSHJ2018-Factorized	0.92	0.94	0.05
BMSHJ2018-Hyperprior	1.13	0.85	0.09
MBT2018-Mean	0.68	0.97	0.02
MBT2018	1.24	0.84	0.23
Cheng2020-Anchor	1.04	0.80	0.20
Cheng2020-Attention	0.99	0.87	0.15
TCM	0.71	0.13	0.78
FTIC	1.11	**0.02**	**0.97**

5.3. Instability in Cheng2020-Anchor

On the $\mathbf{X}_k^{(2)}$ sweep at n=256, we report the unclamped per-probe error $\|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2$ of Eq. (5); inputs have unit energy. The released Cheng2020-Anchor checkpoints diverge at isolated frequencies (Suppl. Fig. S2): the N=128, q=1-3 checkpoints stay below an error of 500, whereas the N=192, q=4-6 checkpoints exceed an error of 10 on 42, 62, and 71 of the 256 frequencies, respectively, reaching 1.9×10^{11}. Because decoder width, bitrate, and training instance change together, we restrict this finding to the released N=192, q=4-6 checkpoint family; architecture and training effects are not identified separately. The $\mathbf{X}_k^{(1)}$ geometry produces the same pattern over broader high-frequency bands (Suppl. Fig. S3). Clamping outputs to $[0, 1]$ bounds the numerical values but does not repair the response: the median per-probe error at q=6 is 0.65, versus 0.05 for the closely related Cheng2020-Attention under the same protocol (Fig. 5). On the basis-matrix probe, Cheng2020-Anchor also becomes non-monotonic for $n \geq 256$ and produces visible artifacts at

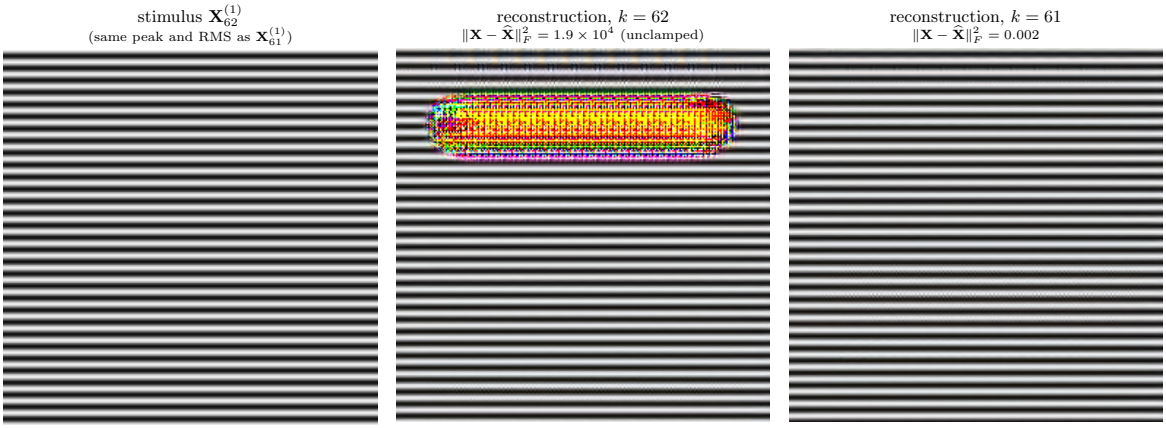
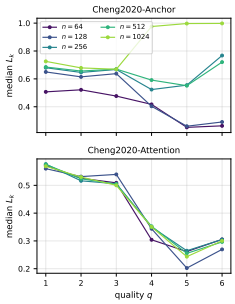
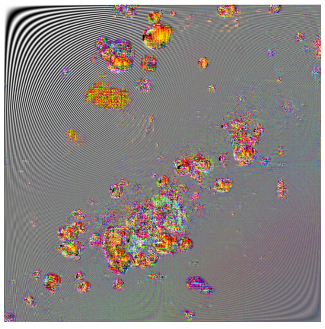


Figure 5. Released-checkpoint instability of Cheng2020-Anchor: clamped reconstructions of neighboring $\mathbf{X}_k^{(1)}$ gratings with matched peak and RMS (n=256, peak 0.45, q=6). The k=62 input produces a localized chromatic artifact (pre-clamp error $1.9{\times}10^4$), whereas k=61 is reconstructed with error 0.002. Full unclamped sweeps: Suppl. Figs. S2 and S3.



(a) Median leakage vs. quality q across DCT image sizes (↓ better).



(b) Cheng2020-Anchor reconstruction at 1024×1024 (q=6).

Figure 6. Size and quality dependence of the composite-basis response. Cheng2020-Anchor becomes non-monotonic for $n{\geq}256$ and produces visible artifacts at 1024×1024.

1024×1024 (Fig. 6). Similar extreme outputs have been reported for these checkpoints under adversarial perturbations [35]; here they are elicited by structured, low-contrast probes.

5.4. Transfer to Natural Images

The controlled probes characterize each codec's spectral response; here we delimit how far these measurements alone extend to natural images, using six Kodak center crops (512×512) and four codecs that span the observed leakage range (Cheng2020-Anchor, Cheng2020-Attention, and BMSHJ2018-Factorized at q=6; JPEG at quality 85).

The prediction target is exact: let $\mathbf{C} = \mathbf{D}^\top\mathbf{X}$ and $\widehat{\mathbf{C}} = \mathbf{D}^\top\widehat{\mathbf{X}}$. Orthonormality of $\mathbf{D}$ gives

$$\|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2 = \sum_{k=0}^{n-1} \underbrace{\|\mathbf{c}_k - \widehat{\mathbf{c}}_k\|^2}_{\delta_k}, \tag{17}$$

where $\mathbf{c}_k$ denotes row k of $\mathbf{C}$ and δ_k is the exact squared-error contribution of frequency k.

Two *input-independent* models, measured once on the basis, predict δ_k without seeing the image: pure attenuation, $\widehat{\mathbf{c}}_k = (1 - \alpha_k)\mathbf{c}_k$ with $\alpha_k{=}1 - \mathcal{D}(\widehat{\mathbf{d}}_k)_k$, and the full linear response $T_{ik} = \mathcal{D}(\widehat{\mathbf{d}}_k)_i$ applied to $\mathbf{C}$. Neither transfers: the attenuation model overpredicts the total squared error by factors of 1.7-10 for the learned codecs and underpredicts JPEG by more than an order of magnitude, and the full operator overpredicts by factors of 8-44. This conclusion is limited to the two models tested here.

The mechanism is content adaptivity: rate and quantization depend on the whole spectrum, so an isolated unit-energy frequency is coded in a different regime than the same frequency embedded in a natural image, and no fixed response measured on one stimulus can represent both. What survives at the codec level is reported in Table 2 as the descriptive statistics $\tilde{\mathcal{L}}$ and $\tilde{\mathcal{L}}/\bar{\rho}$ (Sec. 3.6): computed purely from spectral overlap, they recover the quality ordering of the eight NIC models ($\rho_s{=}{-}0.98$ and -0.93 with PSNR), with the $\tilde{\mathcal{L}}$ ordering identical on all three datasets. The added-distortion control (Suppl. Sec. C.3) reaches similar correlations without the leakage weighting.

HiFiC illustrates the distinction between perceptual quality and target-frequency fidelity: at 0.47 bpp on natural images (basis-probe rate 0.70 bpp; Fig. 2), roughly half the rate of the Table 2 codecs, it attains a lower LPIPS than any of them (0.040), yet the highest normalized alignment ($\tilde{\mathcal{L}}/\bar{\rho}$=0.78) and basis leakage of 0.62-0.77 (Fig. 4).

5.5. Leakage-Regularized Decoder Fine-Tuning

Single-codec demonstration. We fine-tune only the Cheng2020-Anchor decoder on one 128×128 DCT-basis input, using the objective of Eq. (16) with $\mathcal{L}_{\text{dist}}$=0. After 900 steps, median leakage decreases from 0.291 to 0.090 (69%) while low-frequency energy and the bitstream remain unchanged (Fig. 3(c)). Under the standardized 1000-step protocol used below, the same configuration reaches $L^{\text{med}} \approx 0.001$ (Suppl. Table S2). Leakage can thus be minimized on the training basis; a natural-image benefit does not follow.

Basis-only control. Extending this configuration (Eq. (16), $\mathcal{L}_{\text{dist}}$=0) to all eight architectures for 1000 steps, held-out Kodak PSNR decreases by 10.6-22.0 dB for the six older NICs and by 1.7-1.9 dB for TCM and FTIC (Suppl. Table S2). The synthetic

Table 2. Natural-image evaluation of eleven codecs on **115 images** (24 Kodak, 41 CLIC, 50 DIV2K) at per-dataset settings near 0.9 bpp. Learned codecs use fixed settings; classical codecs are rate-selected per dataset. L_k is median leakage on the synthetic 512×512 basis (for classical codecs, the median over three selected settings). $\tilde{\mathcal{L}}$ and $\tilde{\mathcal{L}}/\bar{\rho}$ are descriptive spectral-alignment statistics. Entries with ± report mean and standard deviation over images; best displayed values in directed columns are bold. Rows are ordered by decreasing $\tilde{\mathcal{L}}$.

Codec	$L_k \downarrow$	bpp	PSNR↑ (dB)	MS-SSIM↑	LPIPS↓	$\tilde{\mathcal{L}}$	$\tilde{\mathcal{L}}/\bar{\rho}$
BMSHJ2018-Factorized	0.455	0.98 ± 0.40	35.17 ± 2.77	0.989 ± 0.005	0.087 ± 0.064	0.1486 ± 0.0225	0.6872
Cheng2020-Anchor	0.729	0.82 ± 0.44	36.47 ± 2.43	0.990 ± 0.004	0.074 ± 0.061	0.1372 ± 0.0402	0.7252
Cheng2020-Attention	0.296	0.82 ± 0.44	36.37 ± 2.40	0.990 ± 0.004	0.077 ± 0.063	0.1067 ± 0.0192	0.5576
BMSHJ2018-Hyperprior	0.146	0.96 ± 0.48	36.53 ± 2.03	0.990 ± 0.004	0.082 ± 0.063	0.0825 ± 0.0192	0.4432
MBT2018-Mean	0.116	0.95 ± 0.50	36.71 ± 1.97	0.990 ± 0.004	0.079 ± 0.062	0.0509 ± 0.0099	0.2799
MBT2018	0.115	0.92 ± 0.50	36.92 ± 2.02	0.991 ± 0.004	0.070 ± 0.056	0.0453 ± 0.0088	0.2527
FTIC	0.005	0.89 ± 0.52	37.78 ± 1.90	**0.992 ± 0.004**	0.064 ± 0.057	0.0055 ± 0.0013	0.0347
TCM	**0.002**	0.91 ± 0.52	**37.95 ± 1.87**	**0.992 ± 0.004**	**0.063 ± 0.056**	0.0011 ± 0.0002	0.0069
JPEG	0.022	0.83 ± 0.31	31.39 ± 3.22	0.975 ± 0.009	0.094 ± 0.049	0.0100 ± 0.0011	0.0304
JPEG XL	0.004	0.87 ± 0.32	33.70 ± 3.13	0.983 ± 0.010	0.080 ± 0.037	0.0019 ± 0.0004	0.0077
WebP	0.003	0.92 ± 0.50	34.08 ± 2.56	0.982 ± 0.008	0.076 ± 0.043	0.0004 ± 0.0001	0.0018

NIC-only Spearman (n=8): $\rho_s(\tilde{\mathcal{L}}, \cdot) = -0.98, -0.86, 0.79$ for PSNR, MS-SSIM, and LPIPS, respectively (p=3×10^{-5}, 0.007, 0.021).

Table 3. Leakage-regularized decoder fine-tuning: DCT-basis leakage and Kodak quality, with an MSE-only control. $L^{\text{med}}_{\text{before/after}}$ is the median leakage on the 256×256 composite basis; Δ metrics are reported on Kodak (Δ: fine-tuned minus pretrained) with the bitstream fixed. Accepted operating points minimize L^{med} subject to 0.1 dB validation-PSNR limit. The MSE-only value is selected independently by the same rule. The BMSHJ2018-Factorized step-2000 endpoint exceeds this limit and is shown for completeness. Full protocol: Suppl. Sec. F.1.

	MSE + leakage						MSE-only control
Codec	λ_{freq}	step	$L^{\text{med}}_{\text{before}}$	$L^{\text{med}}_{\text{after}}$	ΔPSNR	ΔLPIPS	$L^{\text{med}}_{\text{after}}$ (ΔPSNR)
BMSHJ2018-Fact.	0.003	2000	0.4558	0.0437	-0.59	−5e−5	0.3812 (-0.02)
BMSHJ2018-Hyp.	0.01	2000	0.1474	0.0034	-0.26	+7e−4	0.1322 (-0.15)
MBT2018-Mean	0.01	1500	0.1244	0.0042	-0.29	+2.3e−3	0.1130 (-0.17)
MBT2018	0.003	2000	0.1151	0.0045	-0.21	+2.4e−3	0.0867 (-0.11)
Cheng2020-Anchor	0.03	500	0.7697	0.0012	-0.17	+6e−4	0.1431 (-0.02)
Cheng2020-Attn.	0.03	1750	0.3023	0.0006	-0.26	+2.9e−3	0.1417 (-0.14)
TCM	0.03	2000	0.0019	0.0001	-0.09	−4e−4	0.0019 (+0.00)
FTIC	0.03	1000	0.0044	0.0001	-0.13	−1e−4	0.0044 (+0.00)

objective alone is insufficient; the protocol below adds natural-image MSE.

Fine-tuning protocol. Table 3 evaluates leakage-regularized decoder fine-tuning across eight NIC architectures. Each fine-tuning step combines pixel MSE on a batch of 4 random crops from DIV2K (256×256) with weighted leakage on the fixed DCT basis matrix (256×256) and two terms that preserve the pretrained low-frequency response, as defined in Eq. (16). Fine-tuned checkpoints are evaluated at 250-step intervals up to step 2,000. Among the evaluated (λ_{freq}, step) pairs, we select the point with the lowest median basis leakage subject to an approximately 0.1 dB PSNR-loss limit on eight held-out DIV2K images. After selection, the chosen decoder is evaluated on Kodak-24 at each codec's native q (rate reported from the entropy model); Table 3 reports mean ΔPSNR and ΔLPIPS (Δ: fine-tuned minus pretrained). Sampling, codec settings, and the evaluated λ_{freq} values are specified in Suppl. Sec. F.1.

Results and control. For BMSHJ2018-Hyperprior, both MBT models, and both Cheng2020 models, leakage-regularized fine-tuning reduces basis leakage to 0.0006-0.0045, compared with 0.087-0.143 for independently selected MSE-only controls; TCM and FTIC both reach approximately 10^{-4}. BMSHJ2018-Factorized also reduces leakage (0.456 → 0.044), but its validation PSNR decreases by 0.416 dB, beyond the 0.1 dB limit. Across the seven models accepted by the validation gate, leakage-regularized fine-tuning reduces median DCT-basis leakage by factors of 19-236 relative to the MSE-only control. On Kodak, their PSNR decreases by at most 0.29 dB; BMSHJ2018-Factorized decreases by 0.59 dB. For the selected Cheng2020 checkpoints, median leakage also decreases on a separate 512×512 basis: 0.73 → 0.03 for Anchor and 0.30 → 0.02 for Attention. Full trajectories are reported in Suppl. Fig. S4.

Despite this small average cost, decoder fine-tuning removes localized chromatic artifacts from the same bitstream in selected held-out examples (Fig. 7; Suppl. Figs. S5-S6).

6. Conclusion

Controlled DCT inputs reveal frequency-dependent codec behavior that aggregate rate-distortion scores do not capture. The basis-matrix probe measures target-frequency gain and off-target redistribution under a joint stress input, while isolated single-frequency probes provide an exact decomposition of target-amplitude error and off-target energy with two-sided error bounds.

Across the evaluated codecs, the older BMSHJ, MBT, and Cheng2020 models show substantially higher high-band off-target energy than the classical codecs, whereas TCM and FTIC fall in the classical-codec range under this diagnostic. The isolated probes also expose a frequency-specific pre-clamp instability in released Cheng2020-Anchor checkpoints that scalar metrics do not reveal.

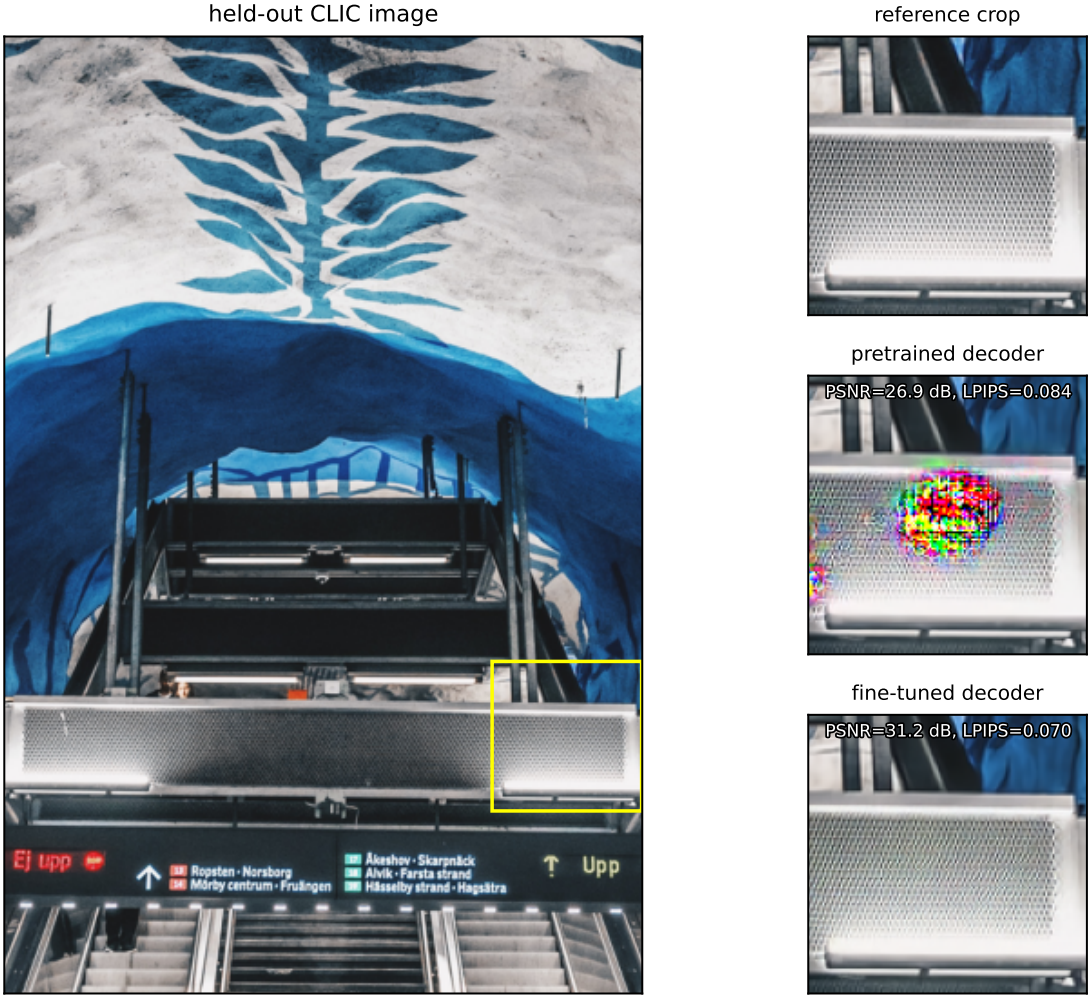


Figure 7. Leakage-regularized decoder fine-tuning on a held-out CLIC image. For Cheng2020-Anchor (q=6), the selected decoder removes the chromatic artifact from the same bitstream and raises PSNR from 26.9 to 31.2 dB.

Relative off-target energy must be interpreted together with target-frequency gain. In the evaluated transfer tests, synthetic response models did not predict natural-image distortion quantitatively, and basis-only fine-tuning did not preserve natural-image quality. The benchmark should therefore be used as a controlled spectral diagnostic alongside standard natural-image evaluation.

The same metric can also be used during decoder fine-tuning to penalize reconstructed energy outside the target frequency. Among the seven accepted models, this penalty reduced off-target energy on the DCT probes by 94.7-99.8% without changing the encoded bitstream. Natural-image PSNR decreased by 0.09-0.29 dB, while selected held-out examples showed removal of localized chromatic artifacts. Extending the analysis to chroma and temporal frequencies requires separate experiments.

CRediT authorship contribution statement

Nikolay I. Kalmykov: Conceptualization, Methodology, Software, Investigation, Visualization, Writing – original draft. **Razan Dibo:** Formal analysis. **Yujun Du:** Formal analysis, Writing – review & editing. **Yipeng Liu:** Supervision, Funding acquisition, Writing – review & editing. **Ivan Oseledets:** Supervision, Funding acquisition. **Anh-Huy Phan:** Conceptualization, Methodology, Validation, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT (OpenAI) in order to polish grammar and phrasing, suggest literature-search keywords, and help debug minor Python issues in scripts. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Supplementary Material

Benchmarking Neural Image Compression with DCT-Based Spectral Diagnostics

Table S1. Core notation used in the main paper and Supplementary Material.

Notation	Definition
$n,\ k$	DCT size and frequency index, $k \in \{0, \dots, n-1\}$.
$\mathbf{D},\ \mathbf{d}_k$	Orthonormal DCT-basis image and its k-th column.
$\mathcal{D},\ \mathcal{D}_2$	Orthonormal one- and two-dimensional DCT operators.
$\widehat{\mathbf{d}}_k,\ \widehat{\mathbf{X}}$	Reconstructed basis-matrix column and isolated probe.
$R_{i,k}$	Fraction of reconstructed column k's DCT energy located in bin i.
$L_k,\ g_k$	Relative off-target energy and signed target-frequency gain.
$\mathbf{X}_k^{(1)},\ \mathbf{X}_k^{(2)}$	Unit-energy isolated probes ($\mathbf{X}$ denotes either); target bins $(k,0)$, (k,k).
s	Amplitude scale of the centered embedding $x \mapsto \frac{1}{2} + sx$.
$E_k^{\text{off}},\ \tau_k,\ \lambda_k$	Off-bin energy, total reconstructed spectral energy, and isolated-probe relative leakage.
$\mu_k, \Delta_k, \sigma_k^{\text{norm}}, H_k^{\text{norm}}$	Spectral centroid, normalized shift, spread, and entropy.
$\text{ODR}_k^{\text{norm}}$	Bounded off-/on-target energy ratio, monotone in L_k.
$\bar{L},\ L^{\text{med}},\ L_{\text{low/high}}$	Mean, median, and low/high-band means of L_k.
$\rho(f),\ \bar{\rho}$	Natural-image radial distortion fraction and its mean over frequency bands.
$\widetilde{\mathcal{L}},\ \widetilde{\mathcal{L}}/\bar{\rho}$	Descriptive synthetic-natural spectral alignment and its magnitude-normalized form.
$\mathbf{c}_k,\ \delta_k$	Row k of the column-DCT coefficients and its exact squared-error share.
$T_{ik},\ \alpha_k$	Measured linear response operator and basis-measured attenuation.
$\mathcal{L}_{\text{freq}},\ \lambda_{\text{freq}}$	Leakage regularizer and its weight in decoder fine-tuning.
$w_k,\ k_0,\ p$	Weights of $\mathcal{L}_{\text{freq}}$, their cutoff index, and exponent.
$\mathcal{L}_{\text{dist}},\ \mathcal{G}_{\text{pur}},\ \mathcal{G}_{\text{mse}}$	Natural-image MSE term; low-band purity/MSE guards.
q, bpp	Checkpoint/quality index and rate in bits per pixel.
ϵ	Numerical constant (10^{-12}).

A. Notation and Definitions

Table S1 collects the notation used throughout the main paper and Supplementary Material.

B. Probe Construction and Mathematical Details

B.1. DCT Preliminaries

Basis. The orthonormal DCT-II basis matrix $\mathbf{D} \in \mathbb{R}^{n\times n}$ has entries

$$\mathbf{D}[m,k] = \alpha_k \cos\Big[\tfrac{\pi}{n}(m+\tfrac{1}{2})k\Big], \qquad \alpha_0 = \sqrt{\tfrac{1}{n}}, \quad \alpha_{k>0} = \sqrt{\tfrac{2}{n}}, \tag{S1}$$

and satisfies $\mathbf{D}^\top\mathbf{D} = \mathbf{I}$. Its k-th column $\mathbf{d}_k$ is the k-th basis wave and has unit norm.

Basis, transform, and probe. The bold $\mathbf{D}$ is the matrix of Eq. (S1), used both as a change of basis and, read as an image, as the basis-matrix probe. The script $\mathcal{D}(\mathbf{v}) = \mathbf{D}^\top\mathbf{v}$ is the one-dimensional transform of a single column, used to build the response matrix R. The script $\mathcal{D}_2(\mathbf{X}) = \mathbf{D}^\top\mathbf{X}\mathbf{D}$ is the two-dimensional transform, i.e. $\mathcal{D}$ applied to every column and then to every row. Orthonormality gives

$$\mathcal{D}_2(\mathbf{D}) = \mathbf{D}^\top\mathbf{D}\,\mathbf{D} = \mathbf{D}, \tag{S2}$$

so the basis-matrix probe is a fixed point of the 2-D DCT: analyzing it in two dimensions returns the probe itself and carries no per-frequency information. The basis-matrix probe is therefore analyzed columnwise with $\mathcal{D}$, while the isolated probes of Sec. B.3, which are single 2-D components by construction, are analyzed with $\mathcal{D}_2$.

Energy. Both transforms are orthonormal, so $\|\mathcal{D}(\mathbf{v})\|_2 = \|\mathbf{v}\|_2$ and $\|\mathcal{D}_2(\mathbf{X})\|_F = \|\mathbf{X}\|_F$ (Parseval). The error decompositions and bounds of Sec. B.4 rest on this equality of spatial and spectral energy.

Index and frequency. Index k corresponds to the spatial frequency $f_k = k/(2n)$ in cycles per pixel: k=0 is the DC term and the Nyquist limit f=0.5 is reached at k=n, so the sampled range $k \le n-1$ spans the full band. This mapping is used by the perceptual-weighting check of Sec. C.1.

B.2. Diagonal of the DCT Basis as a Chirp Signal

Define the diagonal $s_k = \mathbf{D}[k,k]$, $k = 0, \dots, n-1$. Then s_k is a discrete chirp-like signal whose effective instantaneous frequency f_k^{eff} rises and falls periodically, with turning points near $k \approx n/2$, caused by Nyquist folding of its quadratic phase.

Proof. Setting $m = k$ in Eq. (S1) gives $s_k = \alpha_k \cos\Big[\frac{\pi}{n}(k^2 + \frac{k}{2})\Big]$ with phase $\phi_k = \frac{\pi}{n}(k^2 + \frac{k}{2})$. The forward difference yields the instantaneous angular frequency

$$\omega_k = \phi_{k+1} - \phi_k = \frac{\pi}{n}\Big(2k + \tfrac{3}{2}\Big), \quad f_k = \frac{\omega_k}{2\pi} = \frac{2k+1.5}{2n}.$$

Because ϕ_k is quadratic, f_k grows linearly with k, making s_k chirp-like. Whenever f_k exceeds the Nyquist limit 0.5 cycles/sample, the real cosine aliases it; writing $\tilde{f}_k = f_k \bmod 1$ for the aliased frequency,

$$f_k^{\text{eff}} = \min(\tilde{f}_k,\ 1 - \tilde{f}_k) \in [0, \tfrac{1}{2}].$$

Evaluating the analytic phase extension at key indices gives $f_0 = \frac{3}{4n}$, $f_{n/2} = \frac{1}{2} + f_0$, and $f_n = 1 + f_0$; hence the effective frequency approaches its trough at the end of the sampled diagonal, while near $k \approx n/2$ it reaches the Nyquist-folded peak $\frac{1}{2} - f_0$. Between folding points f_k^{eff} rises linearly; after each fold it falls, producing the periodic rise-fall pattern along the diagonal. □

B.3. Single-Frequency Construction Details

Construction. Both probes of Eqs. (3)-(4) are outer products, so their entries are read off directly:

$$\mathbf{X}_k^{(1)}[m,j] = n^{-1/2}\,\mathbf{d}_k[m], \qquad \mathbf{X}_k^{(2)}[m,j] = \mathbf{d}_k[m]\,\mathbf{d}_k[j].$$

Every column of $\mathbf{X}_k^{(1)}$ is the same wave $\mathbf{d}_k$, so the image is constant along rows and varies only down the columns: a horizontal grating. $\mathbf{X}_k^{(2)}$ varies along both axes and forms a checkerboard (Fig. S1). The factor $n^{-1/2}$ normalizes the first probe, since $\|\mathbf{d}_k\mathbf{1}^\top\|_F = \|\mathbf{d}_k\|_2\|\mathbf{1}\|_2 = \sqrt{n}$, whereas $\mathbf{d}_k\mathbf{d}_k^\top$ already has unit norm.

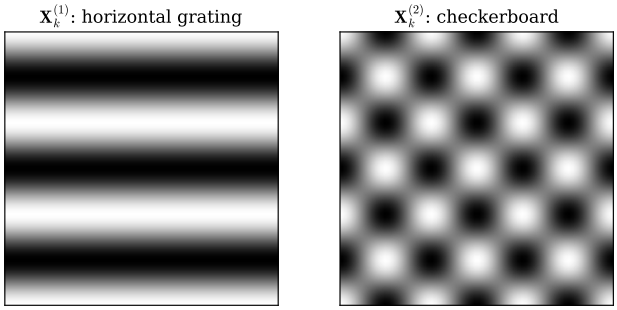

Figure S1. Isolated probes (n=64, k=6). $\mathbf{X}_k^{(1)}$ repeats one basis wave along every column: a horizontal grating with all spectral energy in bin $(k, 0)$. $\mathbf{X}_k^{(2)}$ is separable in both axes: a checkerboard with all energy in bin (k, k).

Amplitude and error scales. The embedded stimulus $\frac{1}{2} + s\mathbf{X}$ is quoted either through its peak amplitude $a = s \max_{m,j} |\mathbf{X}[m, j]|$, where a=0.45 places the stimulus in $[0.05, 0.95]$ around mid-grey, or through the raw scale s applied to the unit-energy probe, whose root-mean-square entry is $1/n$. The two conventions differ by an n-dependent factor and are stated per suite in Sec. F. Unit input energy also fixes the scale of the squared error $e^2 = \|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2$ of Eq. (5): e^2=0 is exact reconstruction, $e^2 \approx 1$ corresponds to discarding the stimulus and returning a flat field, and e^2>1 means the output carries more error energy than the input contained. The divergence threshold used in Sec. C.2 is e^2>10.

Centered embedding check. The isolated probes use $x \mapsto \frac{1}{2} + sx$ with a fixed, k-independent scale. In a controlled simulation, a pure 10% attenuation passed through a per-image min-max map produces a spurious, k-dependent Frobenius error several times the true value, whereas the centered map recovers the injected 0.10 exactly. The centered map is therefore required to keep the stimulus amplitude and DC level independent of frequency.

Single-bin property. For the constructions in Eqs. (3)-(4), $\mathbf{D}^\top \mathbf{d}_k = \mathbf{e}_k$ by orthonormality and $\mathbf{1}^\top \mathbf{D} = \sqrt{n}\, \mathbf{e}_0^\top$ because $\mathbf{d}_0 = n^{-1/2}\mathbf{1}$. Hence

$$\mathcal{D}_2(\mathbf{X}_k^{(1)}) = n^{-1/2}(\mathbf{D}^\top \mathbf{d}_k)(\mathbf{1}^\top \mathbf{D}) = \mathbf{e}_k \mathbf{e}_0^\top, \tag{S3}$$

$$\mathcal{D}_2(\mathbf{X}_k^{(2)}) = (\mathbf{D}^\top \mathbf{d}_k)(\mathbf{d}_k^\top \mathbf{D}) = \mathbf{e}_k \mathbf{e}_k^\top. \tag{S4}$$

Thus each probe carries one unit of energy in exactly one 2-D DCT bin before compression, which makes reconstruction errors comparable across probes and across frequencies.

B.4. Metric Identities and Leakage-Bound Proofs

All exact relationships below were checked on every evaluated codec run; the largest numerical residual is below 10^{-6}.

Attenuation vs. redistribution. Pure attenuation, $\widehat{\mathbf{d}}_k = \alpha \mathbf{d}_k$, gives L_k=0 and g_k=α: leakage alone would report a perfect response. Pure redistribution at preserved target amplitude, $\widehat{\mathbf{d}}_k = \mathbf{d}_k + \sum_{i \neq k} c_i \mathbf{d}_i$, gives g_k=1 and $L_k = \sum_{i \neq k} c_i^2 / (1 + \sum_{i \neq k} c_i^2) > 0$. Neither quantity determines the other, which is why both are reported.

ODR and leakage. Each positive-energy column of R sums to one, so $\sum_{i \neq k} R_{i,k} = L_k$ and

$$\mathrm{ODR}_k^{\mathrm{norm}} = \tanh\left[\frac{L_k}{2(1 - L_k) + 2\epsilon}\right]. \tag{S5}$$

ODR is therefore a strictly monotone reparameterization of leakage (maximum identity residual $9{\cdot}10^{-12}$ across the 11 codecs). In finite precision, saturation of tanh creates ties above $L_k \gtrsim 0.97$.

Pooled auxiliary correlations. Pooled over the eight learned codecs at their nearest-1 bpp configurations of Fig. 2 (2048 frequency-codec pairs), $\rho_s(L_k, H_k^{\mathrm{norm}}) = 0.98$ and $\rho_s(L_k, \sigma_k^{\mathrm{norm}}) = 0.92$. Centroid shift is the most distinct auxiliary descriptor, although it remains correlated with leakage ($\rho_s(L_k, |\Delta_k|) = 0.81$).

Parallel-axis identity. In raw index units, the second moment about the target decomposes as

$$\sum_i (i - k)^2 R_{i,k} = \underbrace{\sum_i (i - \mu_k)^2 R_{i,k}}_{\sigma_k^2} + (\mu_k - k)^2. \tag{S6}$$

Thus σ_k measures symmetric dispersion and $\mu_k - k$ coherent displacement. The normalized σ_k^{norm} and Δ_k of Sec. 3.2 use different denominators, so the exact identity applies to the raw index-domain quantities.

Single-frequency decomposition and bounds. Parseval's identity for the orthonormal 2-D DCT gives

$$\|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2 = (1 - g_k)^2 + E_k^{\mathrm{off}}, \tag{S7}$$

because the input spectrum has one unit entry and all other coefficients are zero. Write $\tau_k = \|\widehat{\mathbf{G}}\|_F^2 = g_k^2 + E_k^{\mathrm{off}} > 0$. Then

$$\|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2 - \lambda_k = \tau_k - 2g_k + \frac{g_k^2}{\tau_k} \tag{S8}$$

$$= \frac{(\tau_k - g_k)^2}{\tau_k} = \frac{(E_k^{\mathrm{off}} - g_k(1 - g_k))^2}{\tau_k} \geq 0. \tag{S9}$$

This proves the lower bound in Eq. (6); equality holds iff $E_k^{\mathrm{off}} = g_k(1 - g_k)$. If $\tau_k = 1$ and $g_k \geq 0$, then $g_k = \sqrt{1 - \lambda_k}$ and

$$\|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2 = 2(1 - \sqrt{1 - \lambda_k}) \leq 2\lambda_k, \tag{S10}$$

which proves Eq. (7). The constants are attained, so they are optimal.

Full-basis bounds. For column k, let $E_k = \|\mathcal{D}(\widehat{\mathbf{d}}_k)\|^2 > 0$ and $a_k = \mathcal{D}(\widehat{\mathbf{d}}_k)_k$, so $a_k^2 = R_{k,k} E_k$. Parseval gives $\|\widehat{\mathbf{d}}_k - \mathbf{d}_k\|^2 = E_k - 2a_k + 1$. If $a_k < 0$, this is at least $1 \geq L_k$. If $a_k \geq 0$, substitute $a_k = \sqrt{R_{k,k} E_k}$; minimizing over $E_k > 0$ gives $E_k = R_{k,k}$ and minimum $1 - R_{k,k} = L_k$. Averaging proves Eq. (8). Under $E_k = 1$ and $a_k \geq 0$,

$$\|\widehat{\mathbf{d}}_k - \mathbf{d}_k\|^2 = 2(1 - \sqrt{R_{k,k}}) \leq 2(1 - R_{k,k}) = 2L_k, \tag{S11}$$

and averaging proves Eq. (9). The energy-preservation condition is substantive: the released Cheng2020-Anchor checkpoints violate it on unstable isolated inputs, so the upper bound is not unconditional.

C. Secondary Diagnostic Checks

C.1. Perceptual-Weighting Robustness Check

To test the equal weighting of frequencies, let $f_k = k/(2n)$ cycles per pixel and let v be the viewing resolution in pixels per degree. We use the Mannos-Sakrison contrast sensitivity function [42],

$$w_k^{\mathrm{CSF}} \propto (0.0192 + 0.114 v f_k) \exp[-(0.114 v f_k)^{1.1}], \qquad \text{(S12)}$$

normalize it so that $\sum_k w_k^{\mathrm{CSF}} = n$, and define $\bar{L}^{\mathrm{CSF}} = n^{-1} \sum_k w_k^{\mathrm{CSF}} L_k$.

At the operating points of Table 2, the rankings for $v \in \{16, 32, 64\}$ px/deg agree closely with the unweighted ranking ($\rho_s = 0.991$ for all three settings). Applying the same weights inside the natural-image alignment statistic leaves the eight-NIC correlations unchanged ($\rho_s = -0.98$ with PSNR and 0.79 with LPIPS). The tested weighting thus rescales the profiles without reordering the codecs.

C.2. Instability Sweeps and Gallery

Figure S2 shows the unclamped $\mathbf{X}_k^{(2)}$ sweep underlying Sec. 5.3. Complementing it, we sweep the grating stimuli $\mathbf{X}_k^{(1)}$ through Cheng2020-Anchor without output clamping (n=256, centered embedding $x \mapsto \frac{1}{2} + sx$ with $s = an/\sqrt{2}$, i.e. peak amplitude $a \in \{0.225, 0.45\}$, matched to the half/full-contrast $\mathbf{X}_k^{(2)}$ sweeps). The gratings produce extreme pre-clamp errors at more frequencies than the two-dimensional stimuli: 141/145/147 of the 256 frequencies exceed e^2>10 at q=4/5/6 at full amplitude (36/24/49 at half), and the unclamped error spans roughly 35 orders of magnitude across the sweep; Fig. S3 shows stimulus/reconstruction pairs.

Divergence is not a smooth function of frequency: k=61 is reconstructed near-perfectly while the neighboring k=62 and k=63 diverge, and stable indices (k=70, k=133) survive inside otherwise unstable bands. At moderate divergence the artifact is a localized chromatic region; at extreme divergence (k=169, k=253) the reconstruction is corrupted across the full image. Because the divergent set shrinks at half amplitude, reported instability sets must include the probe amplitude.

C.3. Alignment Behavior Across Added Distortions

We apply four additional distortions: Gaussian noise ($\sigma \in \{0.01, \ldots, 0.2\}$), pure noise, bit-depth quantization (2-6 bits), and JPEG re-compression ($q \in \{10, \ldots, 70\}$) to check whether the L_k-weighted statistic remains monotonic. Since L_k is fixed per codec, only $\rho(f)$ changes. The evaluated distortions follow a broadly monotonic $\tilde{\mathcal{L}}$-PSNR trend; an unweighted $\bar{\rho}$ correlates marginally better ($\rho_s = -0.964$ vs. -0.945, pooled over the eight NIC codecs), so the leakage weighting does not improve the prediction here. We report $\tilde{\mathcal{L}}/\bar{\rho}$ as a descriptive spectral-shape summary, not a predictive metric.

D. Fine-Tuning Diagnostics and Ablations

Figure S4 shows the leakage-PSNR selection trajectories at the λ_{freq} of Table 3. Table S2 reports the basis-only negative control.

Table S2. Basis-only decoder fine-tuning as a negative control. The decoder is optimized on one fixed DCT-basis input (128×128; TCM and FTIC: 256×256) without natural images. $L_{\mathrm{train}}^{\mathrm{med}}$ is the median leakage on that same training probe; ΔPSNR is fine-tuned minus pretrained performance on held-out Kodak images. The bitrate is unchanged because the encoder and entropy model remain frozen.

Codec	DCT size	$L_{\mathrm{train}}^{\mathrm{med}}$ before	$L_{\mathrm{train}}^{\mathrm{med}}$ after	ΔPSNR (dB)
BMSHJ2018-Fact.	128×128	0.445	0.008	-22.0
BMSHJ2018-Hyp.	128×128	0.163	0.003	-13.9
MBT2018-Mean	128×128	0.148	0.002	-16.5
MBT2018	128×128	0.125	0.002	-12.0
Cheng2020-Anchor	128×128	0.287	< 0.001	-10.6
Cheng2020-Attn.	128×128	0.269	< 0.001	-11.5
TCM	256×256	0.002	< 0.001	-1.7
FTIC	256×256	0.004	< 0.001	-1.9

E. Selected Natural-Image Examples

E.1. Failure and Contrast Cases at Nearby Bitrates

Figures S5-S6 show six cases retained after screening Urban100 and the Describable Textures Dataset (5,740 images in total) with the benchmark pipeline. Five rows cover distinct pronounced failures (facade slats, lace, parallel lines, zigzags, and a mesh); the irregular cracked texture is included as a minimally affected contrast. The selection illustrates the range of observed behavior, not its prevalence. Each row shows the original crop, Cheng2020-Anchor at q=6, the same model after the decoder fine-tuning of Sec. 5.5, HiFiC at its highest released rate, and FTIC and JPEG XL at the settings nearest the Cheng bitrate. Crops are deterministic: the 512×512 window of maximum high-frequency energy, or the largest 64-multiple center crop for smaller images. Overlaid on each reconstruction: bitrate, PSNR, MS-SSIM, VIF, LPIPS, HF-PSNR (PSNR restricted to the high band, radial frequency above half-Nyquist), and ρ_{hf} (the distortion fraction $\rho(f)$ of Sec. 3.6 integrated over the high band).

On the affected near-periodic examples, Cheng2020-Anchor produces localized chromatic artifacts and severe quality loss (PSNR 11-18 dB on the lace, line, and facade rows). These artifacts are visually related to the single-frequency failure of Sec. 5.3, but the experiment does not establish a shared mechanism. The fine-tuned decoder removes most of the visible artifacts from the same bitstream, recovering 3-16 dB on the affected rows (facade 17.6 → 26.4 dB, lined 11.5 → 27.6 dB) at zero rate cost. FTIC and JPEG XL show no comparable artifacts at the nearest available bitrates. HiFiC is not rate-matched in every row; its comparison is therefore descriptive.

F. Reproducibility

Source code, evaluation scripts, pre-computed results, notebooks that regenerate every figure, and an installable package are available at `https://github.com/nkalmykovsk/dct_benchmark_nic`, with pinned dependency versions. This section states the settings needed to read the results; the repository holds the remaining implementation detail, including the mapping from each figure and table to the script that produces it.

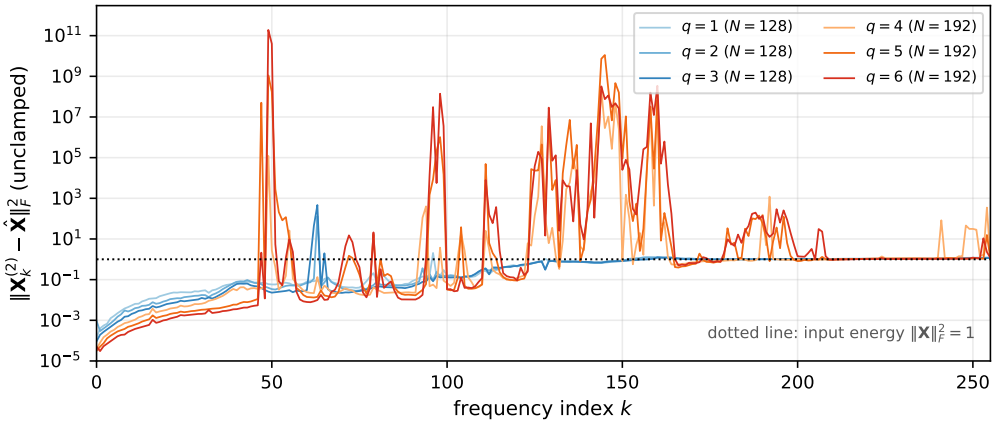

Figure S2. Unclamped squared reconstruction error on the $\mathbf{X}_k^{(2)}$ sweep (n=256, peak 0.45) for the six released Cheng2020-Anchor checkpoints. The N=128, q=1-3 checkpoints have maximum errors below 500; the N=192, q=4-6 checkpoints reach 1.9×10^{11} at isolated indices (dotted line: unit input energy).

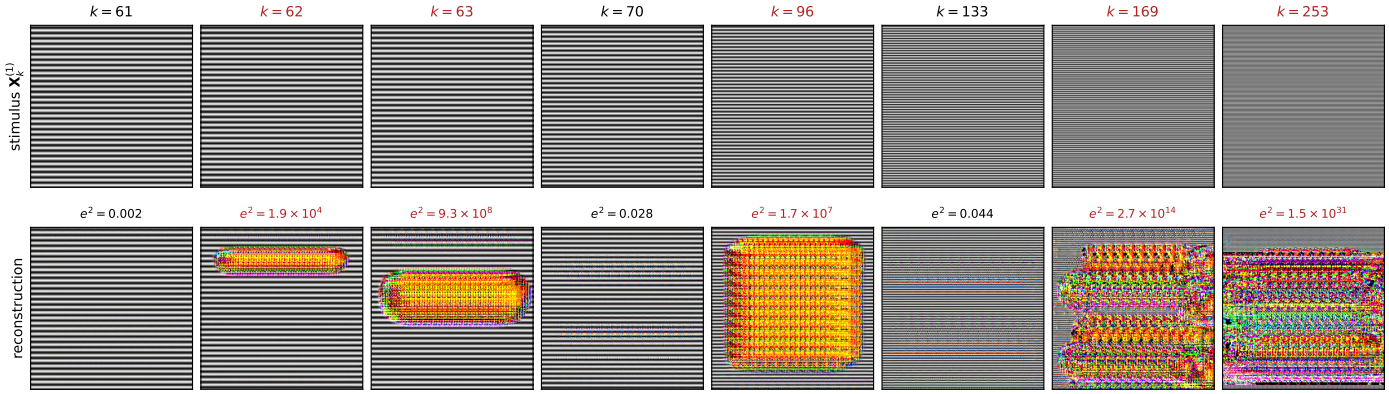

Figure S3. Instability of the grating probe $\mathbf{X}_k^{(1)}$ (n=256, peak amplitude 0.45): selected stimuli and their Cheng2020-Anchor q=6 reconstructions (clamped to [0, 1] for display). Red titles mark frequencies whose unclamped squared error exceeds 10; artifacts range from a localized chromatic region (k=62) to global collapse (k=253), with interleaved stable indices.

Environment. Benchmark experiments use Python 3.11 and PyTorch 2.9 on one NVIDIA A100 (40 GB); extended experiments use Python 3.9 and PyTorch 2.5.1 on one NVIDIA H100 (80 GB). Neural codecs are taken from CompressAI 1.2.8 (BMSHJ and MBT families, Cheng2020-Anchor, Cheng2020-Attention) and from the reference implementations of TCM and FTIC; classical codecs from Pillow (JPEG, WebP) and imagecodecs (JPEG XL); metrics from the reference LPIPS and MS-SSIM implementations.

Determinism. The basis D=DCT(I_n) is deterministic, and all codec weights are frozen pre-trained checkpoints evaluated in inference mode. The basis-only control was repeated with three seeds (leakage standard deviation <0.001); joint fine-tuning uses a single fixed seed, with no multi-seed stability claim.

Codec configurations. CompressAI models and FTIC are evaluated at quality levels $q\in\{1,\dots,6\}$, TCM at $\lambda\in\{0.0025, 0.05\}$ with $p\in\{64, 128\}$, JPEG and WebP at qualities $\{20, 40, 55, 70, 85, 95\}$, and JPEG XL at distances $\{4.0, 2.0, 1.0, 0.6, 0.3, 0.1\}$. DCT bases use $n\in\{64, 128, 256, 512, 1024\}$, restricted to $n\geq256$ for FTIC and TCM.

Bitrate. For the neural codecs, bpp is the entropy-model estimate, i.e. the cross-entropy of the quantized latents under the learned likelihoods, which is the standard CompressAI convention and a tight proxy for the arithmetic-coded length. For JPEG, WebP, and JPEG XL, bpp is the encoded file size divided by the pixel count. Rate-matched comparisons therefore compare an estimated rate against a measured one.

Table 2. 115 images: 24 Kodak at native resolution, 41 CLIC Pro validation, and 50 DIV2K (evenly subsampled), the latter two center-cropped to ≤768 px. NICs at fixed quality q=6; classical codecs at the quality whose mean bitrate is nearest 0.9 bpp per dataset. L_k is computed at the same settings on a 512×512 basis; N_b=512 radial bins. PSNR, MS-SSIM, LPIPS computed on RGB; Spearman correlations over 8 NICs.

Numerical caveat. TF32 must be disabled on Ampere and Hopper GPUs: with TF32 enabled, the MS-SSIM implementation returns invalid values (> 1).

F.1. Fine-Tuning Protocols

Basis-only controls. The basis-only experiment of main-paper Fig. 3(c) updates only the synthesis transform g_s with

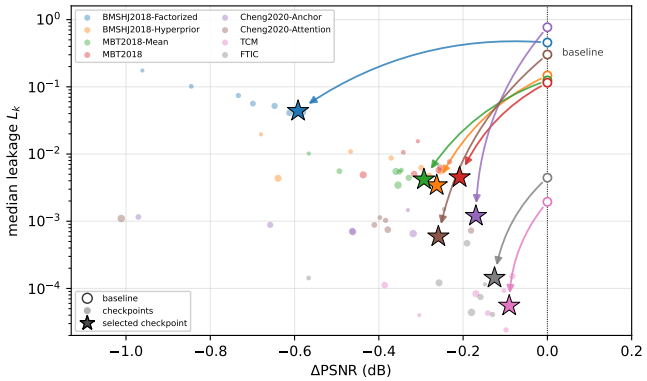

Figure S4. Leakage-regularized decoder fine-tuning trajectories at the λ_{freq} used in Table 3: median DCT-basis leakage (256×256) vs. ΔPSNR on the Kodak images. Small circles show checkpoints evaluated every 250 steps; arrows connect each pretrained decoder to its selected checkpoint (star). For the seven codecs accepted under the −0.1 dB validation limit, leakage decreases by 94.7-99.8% and Kodak PSNR decreases by at most 0.29 dB. BMSHJ2018-Factorized reduces leakage by 90%, but has $\Delta\text{PSNR}_{\text{val}} = -0.416$ dB.

Adam (lr=10^{-4}) for 900 steps on one DCT-basis image (128×128). The objective sets $\mathcal{L}_{\text{dist}}$=0 and λ_{freq}=1 and uses p=2; thus, the normalized weights $w_k \propto (k/(n-1))^2 + 0.2$ are zero for $k < \lfloor 0.33n \rfloor$. It also includes the purity-deficit and low-band MSE terms of Sec. 3.7. Table S2 uses the same decoder-only objective for 1000 steps and three seeds on a 128×128 basis (TCM and FTIC: 256×256).

Table 3: data and updates. Eight DIV2K images are the held-out validation set; the fine-tuning set is the first 400 remaining images in sorted order. Each step samples four images with replacement and draws one random 256×256 crop from each (8000 crops per 2000-step run) and processes one fixed 256×256 DCT basis ($\mathcal{L}_{\text{freq}}$ with p=3, both guards). All runs use Adam (10^{-4}), gradient-norm clipping at 1, and a single fixed seed; leakage-regularized and MSE-only runs share the fine-tuning set but use independent minibatch sequences. The MSE-only control sets λ_{freq}=0 with both guards disabled.

Checkpoint grid. Checkpoints are evaluated at steps $\{0, 250, \ldots, 2000\}$ and selected by the validation rule of Sec. 5.5. We evaluate λ_{freq}=0.03 for all models and additionally 0.01 and 0.003 for the four BMSHJ/MBT models; BMSHJ2018-Factorized has no nonzero-step checkpoint that passes the gate, and Table 3 reports its λ_{freq}=0.003, step-2000 endpoint for completeness. CLIC is used for the held-out example in Fig. 7 and Figs. S5-S6.

F.2. Probe Amplitudes

The two amplitude conventions are not interchangeable (Sec. B.3). The single-frequency suite embeds $\mathbf{X}_k^{(1)}$ and $\mathbf{X}_k^{(2)}$ by $x \mapsto \frac{1}{2} + sx$ over all k, with s=0.225n for $\mathbf{X}^{(2)}$ and s=0.45 $\sqrt{n}$ for $\mathbf{X}^{(1)}$ at n=256, clamped to $[0, 1]$ unless stated. The grating-instability runs (Fig. 5, Fig. S3) instead use $s = an/\sqrt{2}$ with peak amplitude $a \in \{0.225, 0.45\}$ and no clamping; reported instability sets are valid only for the stated amplitude. The profile sweep covers 124 configurations.

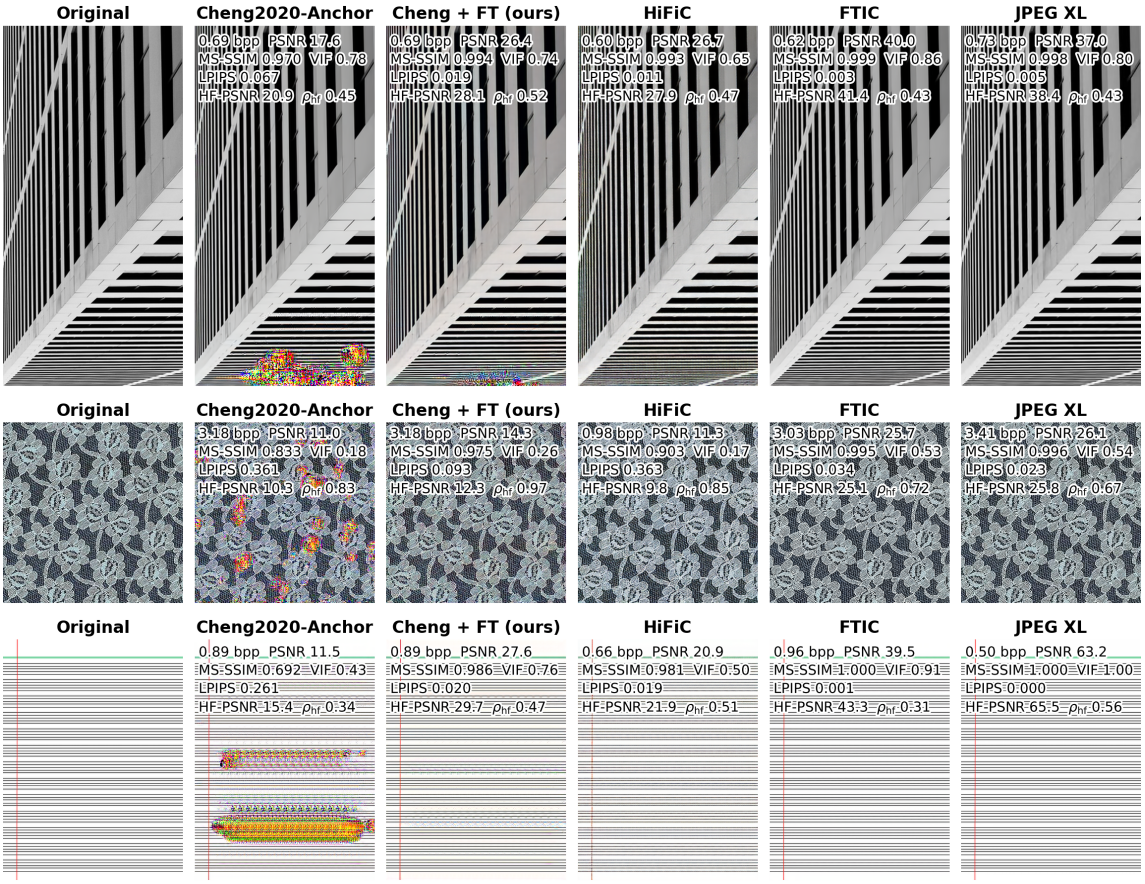

Figure S5. Affected near-periodic cases: Urban100 facade, DTD lace, and DTD parallel lines. Cheng2020-Anchor (q=6) produces localized chromatic artifacts; the fine-tuned decoder processes the same bitstream. FTIC and JPEG XL are shown at the available settings nearest the Cheng bitrate.

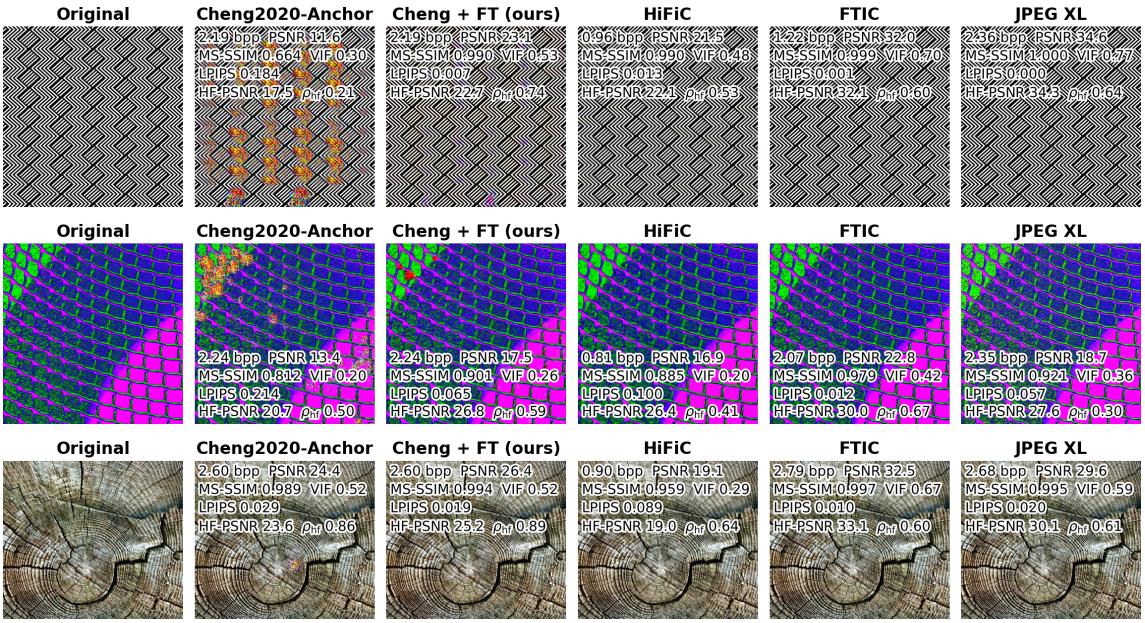

Figure S6. Two affected periodic textures (DTD zigzagged and meshed) and one minimally affected irregular texture (DTD cracked). The final row is a contrast case: high-frequency content alone does not imply a severe failure. Layout and overlays follow Fig. S5.